

Figure 7: The layout manager

Eclipse.

Now that we know how to run a simulator, we will go through the basics of an Android Project in the Eclipse IDE. Figure 6 shows the project hierarchy.

It contains the *src* directory, the Android library, the resource bundle directory and the Android manifest.

Most of the work that concerns the UI of the application is contained in the resource bundle folder called *res*. It has three sub-directories: *drawable* contains all pictures/icons/photos; *layout* has the layout of the application screens; and *values* holds the language file called *strings.xml*, a colour definition file called *colors.xml*, and *styles.xml*, which contains all the styles of the UI components on the screen.

The *AndroidManifest.xml* is the manifest file or something equivalent to a deployment descriptor, which has information about the application -- both descriptive and security information.

The source folder has all the sources related to the activities, viz., events on what action will be taken when buttons are clicked, etc.

Building the application

First, to build a dialler we need to create a dial pad and a place-holder for the numbers in it. Also, we have to place two buttons that will represent the *dial* and the *cancel* activities. We will group both these events under one activity called 'makeacall'.

Now let's build the UI with the layout manager. If you open the *main.xml* file within the *res/layout* directory in the project, it will automatically open the layout manager. There should be a visual representation of the UI as shown in Figure 7.

Just like AWT or Swing in Java, Android also has something called different layouts. Here, the most relevant layout for a dial pad is a *TableLayout*. As per the diagram, the place holder for the number is an *EditText* component and there are four rows represented by a layout called *TableRow*. Each row has four buttons represented by a component called

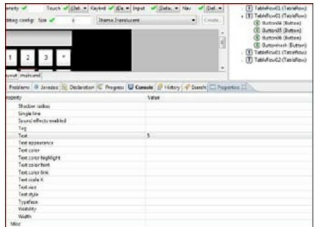


Figure 8: Setting values for the different objects



Figure 9: The dialler

Button. The last row contains two *ImageButtons*, which represent the *dial* and *cancel* buttons.

Once the layout is done, we will configure a few properties for these objects (Figure 8), beautify the contents of the buttons and make them look like a real dial pad.

Once all the properties are configured, the dialler will look something like the image in Figure 9.

The last step is to write the activities. Our activity 'makeacall' will look like the code presented in Listing 1.

Listing 1: Code for 'makeacall' activity

```
package com.lfymag.androidsample;
```

```
import android.app.Activity;
import android.content.ActivityNotFoundException;
import android.content.ContentUri;
import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
```